CONSTRUCTION PROGRAMMES

TEVET QUALIFICATIONS FRAMEWORK (TQF)

CURRICULUM CHART

FOR

CRAFT CERTIFICATE

IN

BRICKLAYING AND PLASTERING

CURRICULUM CHART NO. 74

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1.0 ACKNOWLEDGEMENTS

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2.0 RATIONALE

Zambia has gone through rapid social, economic, cultural and political changes. This has translated booming construction where more housing units schools, hospitals, among others.

In view of the above the Craft Certificate in Bricklaying and Plastering programme has been reviewed to contribute to the improvement in the levels of service, quality, productivity and income generating capacities at a time when the 'never beens' , and those coming out of literacy classes have lost hope of acquiring any form of training to better their lives.

On completion of this programme the trainees will be able to work in the Construction industry either formally or informally.

Through this programme the nation will be provided with a competent human resource which will contribute effectively to its development.

3. 0 PROGRAMME PURPOSE

The purpose of the programme is to equip the trainees with knowledge, skills and appropriate attitudes to work through hands on experiences, supervise and motivate team members, mentor subordinates, and build various structures, fit door and window frames and apply finishes.

4.0 PROGRAMME OUTCOMES

On completion of the programme, trainees will be able to:

1. Practise occupational health and safety in the workplace
2. Use tools, equipment and materials appropriately
3. Fit door and window frames according to specifications
4. Construct various structures
5. Carry out renovations and alterations to structures
6. Install drainage systems
7. Carryout wall plastering, jointing and pointing according to required standards.
8. Communicate effectively and efficiently in the workplace
9. Supervise and motivate team members

10. Apply entrepreneurial skills in bricklaying and plastering
11. Apply mathematical skills in bricklaying and plastering

5.0 DURATION

The course will be for a duration of 2400 notional learning hours or two (2) years inclusive of **480** hours or three (3) months industrial attachment.

6.0 COURSE OUTLINE

Module No.	Title	No. of Hours
Year 1		
74-01-A	Occupational Safety Health and Environment	140
74-02-A	Trade Tools, Equipment and Materials	100
74-03-A	Constructing Structures	720
74-04-A	Communications skills	80
74-05-A	Introduction to Computers	80
74-06-A	Trade Mathematics	160
74-07-A	Door and Window Openings	100
	Sub-total	**1 380**
Year 2		
	Industrial Attachment	480
74-08-B	Finishes	100
74-09-B	Renovating and Altering structures	120
74-10-B	Drainage	100
74-11-B	Entrepreneurship	80
74-12-B	Stairways, Masonry and Fireplaces	100
	Sub-total	**1020**
	Grand Total:	**2 400**

7. 0 TEACHING STRATEGIES

11.1 Practical
11.2 Group discussions
11.3 Demonstration
11.4 Individual projects
11.5 Simulations
11.6 Hand-outs
11.7 Training aids (models, pictures, videos, charts)

8.0 PROGRAMME EVALUATION

TEVETA shall evaluate the programme as follows:

8.1 Formative evaluation

8.1.1 Purpose: To determine on an on-going basis whether the programme is being implemented as planned and to advise on improvements.

8.1.2 Major areas of Evaluation: Course aims and objectives student's entry requirements, course contents, teaching, learning activities, learning resources.

8.1.3 Evaluation Instruments: Questionnaires, structured interviews, observations, checklist, examinations/test records, participation and attendance in class.

8.1.4 Sources of Information: Students, trainers, administrative monitors and records.

8.2 Summative Evaluation

8.2.1 Major areas of evaluation – Course aims, trainees' entry requirements, course content, learning resources, teaching/learning activities, trainers, and graduates performance in employment.

8.2.2 Evaluation Instruments:- Questionnaires, structured interviews, observations, checklists, records, final Examinations.

8.2.3 Sources of information:- Trainees, trainers, administrators, student dropouts, employers, union official and general public.

9.0 ENTRY REQUIREMENTS

9.1 Direct Entry: Five (5) passes including English, Mathematics , Science and any other two (2) subjects.

9.2 Mature Age:

9.2.1 Grade 12 General Certificate of Education

9.2.2 Level I Trade Test Certificate in Bricklaying and Plastering

9.2.3 At least one year relevant industrial experience

10.0 ASSESSMENT

10.1	Final Assessment	100%
10.2	Continuous Assessment	40%
10.3	Pass Mark	50%

11.0 ATTENDANCE

The candidate must have an attendance of minimum 85% and syllabus coverage of 85% to be eligible to sit for the final examinations.

12.0 PROGRESSION REQUIREMENTS

12.1 Candidates will be allowed to write final examination upon passing summative assessment

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12.2 Candidates who have undergone informal training shall be allowed to sit for the final assessment as long as they show a portfolio of prior learning.

13.0 STAFF QUALIFICATION

13.1 Minimum of Craft Certificate in Bricklaying and Plastering or equivalent

13.2 Minimum of five years relevant work experience

13.3 Teaching qualification from an accredited and registered institute

13.4 TEVETA accredited trainer

14.0 CERTIFICATION

Successful trainees will be awarded a Craft Certificate in Bricklaying and Plastering by TEVETA

MODULE: 74-01-A OCCUPATIONAL SAFETY, HEALTH AND ENVIRONMENT
NOMINAL DURATION: 140 HOURS CREDITS: 14
MODULE PURPOSE: To equip the trainees with knowledge, skills and appropriate attitudes to occupational safety, health and environment.
LEARNING OUTCOMES: On completion of this module, trainees will be able to:

1. Apply construction safety standards
2. Observe safety and health in workplace
3. Apply environmental protection measures
4. Conducting local risk control
5. Erect scaffolds
6. Apply green technology in building.

UNIT A1.1 APPLYING CONSTRUCTION SAFETY STANDARDS

1.1.1 Legal Requirements
1.1.2 Statutory Requirements
1.1.3 Environmental Regulation

UNIT A1.2 OBSERVING SAFETY & HEALTH IN THE WORKPLACE

1.2.1 House keeping
1.2.2 Common accidents
1.2.3 Common injuries
1.2.4 First Aid
1.2.5 Health risks
1.2.6 Health Practices

UNIT A1.3 APPLYING ENVIRONMENTAL PROTECTION MEASURES

1.3.1 Solid Waste
1.3.2 Toxic Waste
1.3.3 Sanitation
1.3.4 Personal Protective Equipment

UNIT A1.4 CONDUCTING LOCAL RISK CONTROL

1.4.1 Hazards
1.4.2 Risk Assessment
1.4.3 Risk Treatment
1.4.4 Records and Reports

UNIT A1.5 ERECTING SCAFFOLDS

1.5.1 Materials and accessories
1.5.2 Rules and Regulations in erecting
1.5.3 Safety Measures

1.5.4 Assembling and disassembling

UNIT A1.6 APPLYING GREEN TECHNOLOGY IN BUILDING.

1.6.1 Renewable Energy and Energy Conservation

1.6.2 Water purification

1.6.3 Air purification

1.6.4 Sewerage treatment

1.6.5 Solid waste management

MODULE: 74-02-A TRADE TOOLS, EQUIPMENT AND MATERIALS
NOMINAL DURATION: 100 HOURS CREDIT: 10
MODULE PURPOSE: To equip the trainee with knowledge, skills and appropriate attitudes to effectively and efficiently handle trade tools, equipment and materials.
LEARNING OUTCOMES: On completion of this module, trainees will be able to:

1. Use bricklaying and plastering tools
2. Use bricklaying and plastering equipment
3. Discuss properties of materials
4. Use building materials
5. Care for tools, equipment and materials

UNIT A2.1 USING BRICKLAYING AND PLASTERING TOOLS

2.1.1 Types of hand tools
2.1.2 Methods of handling hand tools
2.1.3 Storage of hand tools

UNIT A2.2 USING BRICKLAYING AND PLASTERING EQUIPMENT

2.2.1 Types of equipment
2.2.2 Usage of various types of equipment
2.2.3 Maintenance of equipment

UNIT A2.3 DISCUSSING PROPERTIES OF MATERIALS

2.3.1 Aggregates
2.3.2 Matrices
2.3.3 Additives
2.3.4 Metals
2.3.5 Walling units

UNIT A2.4 USING BUIDLING MATERIALS

2.4.1 Types of materials
2.3.2 Usage of various types of materials
2.3.3 Storage of materials

UNIT A2.5 CARING FOR TOOLS, EQUIPMENT AND MATERIALS

2.5.1 Cleaning
2.5.2 Sharpening
2.5.3 Checking for accuracy
2.5.4 Storing

UNIT A2.6 **ESTIMATING AND ORDER MATERIALS**

2.6.1 Explain the measuring and ordering of building materials
2.6.2 Select necessary materials
2.6.3 Order necessary materials and quantities
2.6.4 Identify tools and equipment

MODULE: 74-03-A CONSTRUCTING STRUCTURES

NOMINAL DURATION: 720 HOURS CREDITS: 72

MODULE PURPOSE: To equip the trainee with knowledge, skills and appropriate attitudes to effectively and efficiently construct structures.

LEARNING OUTCOMES: On completion of this module, trainees will be able to:

1. Observe general construction and building legislation
2. Interpret building drawings
3. Set out buildings
4. Prepare mortar
5. Cast concrete
6. Build various structures
7. Construct pavements and walkways
8. Construct decorative structures

UNIT A3.1 OBSERVING GENERAL CONSTRUCTION AND BUILDING LEGISLATION

3.1.1 Construction Industrial laws and Regulations
3.1.2 The Building team
3.1.3 Building standards and code of practice
3.1.4 Statutory Instruments

UNIT A3.2 INTERPRETING BUILDING DRAWINGS

3.2.1 Units of measurements
3.2.2 Plans
3.2.3 Elevations
3.2.4 Scaling
3.2.5 Bill of Quantities

UNIT A3.3 SETTING OUT BUILDINGS

3.3.1 Foundations
3.3.2 Cleaning and levelling site
3.3.3 Making profiles

UNIT A3.4 PREPARING MORTAR

3.4.1 Types of mortar
3.4.2 Characteristics
3.4.3 Composition
3.4.4 Ratios
3.4.5 Mixing procedures

UNIT A3.5 CASTING CONCRETE

3.5.1 Types of concrete
3.5.2 Composition and Ratios

3.5.3 Mixing Procedures

3.5.4 Reinforcements

3.5.5 Placing, compacting and curing

UNIT A3.6 BUILDING VARIOUS STRUCTURES

3.6.1 Building principles

3.6.2 Types of bonds

3.6.3 Types of walls

3.6.4 Corners and curves

UNIT A3.7 CONSTRUCTING PAVEMENTS AND WALKWAYS

3.7.1 Surface preparation

3.7.2 Types of pavers and kerbs stones

3.7.3 Setting out

3.7.4 Laying procedures

UNIT A3.8 CONSTRUCTING DECORATIVE STRUCTURES

3.8.1 Decorative brickwork

3.8.2 Arches

3.8.3 Twisted Piers

3.8.4 Methods of construction

3.8.5 Tilling

UNIT A3.9 CONSTRUCTING PILLARS

3.8.1 Types of pillars

3.8.2 Methods of constructions

3.8.3 Uses of pillars

3.8.4 Methods of termination

MODULE: 74-04-A COMMUNICATION SKILLS
NOMINAL DURATION: 80 HOURS CREDITS: 8

MODULE PURPOSE: To equip the trainee with knowledge, skills and appropriate to enable them communicate effectively and efficiently in the workplace.

LEARNING OUTCOMES: On completion of the module, the trainees will be able to:

1. Apply the concepts of effective communication
2. Differentiate types of communication
3. Correspond with stakeholders
4. Write reports
5. Use the Library
6. Apply study skills
7. Apply business correspondence skills
8. Use Technical English

UNIT A4.1 APPLYING THE CONCEPTS OF EFFECTIVE COMMUNICATION

4.1.1 Nature and purpose of communication
4.1.2 Communication process
4.1.3 Barriers to effective communication
4.1.4 Strategies of effective communication

UNIT A4.2 DIFFERENTIATING TYPES OF COMMUNICATION

4.2.1 Verbal / Oral communication
4.2.2 Non – verbal communication
4.2.3 Written communication

UNIT B4.3 CORRESPONDING WITH STAKEHOLDERS

4.3.1 Letters
4.3.2 Memorandum
4.3.3 Circulars
4.3.4 Notices
4.3.5 Forms and documentation (personal/official)

UNIT B4.4 WRITING REPORTS

4.4.1 Types of reports
4.4.2 Report formats
4.4.3 Composing different types of reports

UNIT B4.5 **USING THE LIBRARY**

4.5.1 Filling systems
4.5.2 References and dictionaries
4.5.3 Periodicals
4.5.4 Bibliography
4.5.5 Effective use of library facilities

UNIT B4.6 **APPLYING STUDY SKILLS**

4.6.1 Reading
4.6.2 Writing skills
4.6.3 Listening and speaking

UNIT B4.7 **APPLYING BUSINESS CORRESPONDENCE SKILLS**

4.7.1 Letters
4.7.2 Memoranda
4.7.3 Circulars
4.7.4 Forms and documentation

UNIT B4.8 **USING TECHNICAL ENGLISH**

4.8.1 Technical definitions
4.8.2 Technical instructions
4.8.3 Technical descriptions

MODULE 74-05-A INTRODUCTION TO COMPUTERS

DURATION: 80 HOURS CREDITS: 8

MODULE PURPOSE: To equip the trainees with knowledge, skills and appropriate attitudes in effective and efficiently use of computer applications.

LEARNING OUTCOMES: On completion of this module, the trainees will be able to:

1. Operate a computer system
2. Use an operating system
3. Use word processing
4. Use spread sheet
5. Use the internet
6. Make a power-point presentation

UNIT A5.1 OPERATING A COMPUTER SYSTEM

5.1.1 Switching on and off the computer
5.1.2 Components of a computer
5.1.3 Features of a keyboard and a printer
5.1.4 Different storage devices
5.1.5 State of the Computer

UNIT A5.2 USING AN OPERATING SYSTEM

5.2.1 File system
5.2.2 Libraries
5.2.3 Install programmes
5.2.4 Virus, protection and removal

UNIT A5.3 USING WORD PROCESSING

5.3.1 Document handling
5.3.2 Basic editing functions
5.3.3 Commands

UNIT A5.4 USING SPREADSHEETS

5.4.1 Spread sheet handling
5.4.2 Use of commands
5.4.3 Functions
5.4.4 Charts
5.4.5 Graphs

UNIT A5.5 USING THE INTERNET

5.5.1 The benefits of using the Internet
5.5.2 Browsing the World Wide Web (www)
5.5.3 Saving web pages

5.5.4 Downloading documents from the Web
5.5.5 Rules for using the Internet

UNIT A5.6 MAKING A POWERPOINT PRESENTATION

5.6.1 Presentation handling
5.6.2 Inserts
5.6.3 Formatting
5.6.4 Diagrams and drawings

MODULE: 74-06-A TRADE MATHEMATICS

NOMINAL DURATION 160 HOURS CREDITS: 16

MODULE PURPOSE: To equip trainees with knowledge, skills and required attitudes to effectively and efficiently apply trade mathematics.

LEARNING OUTCOME: On completion of the module, the trainees will be able to:

1. Use numerical methods to solve problems
2. Solve quadratic equations
3. Use approximations/errors
4. Compute arithmetic operations
5. Solve problems using mensuration

UNIT A6.1 USING NUMERICAL METHODS TO SOLVE PROBLEMS

6.1.1 Evaluation of bricklaying and plastering problems
6.1.2 Matrices and determinants
6.1.3 Exponential functions

UNIT A6.2 SOLVING QUADRATIC EQUATIONS

6.2.1 Discriminant
6.2.2 Intercept
6.2.3 Maximum and minimum parts

UNIT A6.3 USING APPROXIMATIONS/ERRORS

6.3.1 Significant figure and decimal places
6.3.2 Standard form (scientific notation)
6.3.3 Indices
6.3.4 Errors

UNIT A6.4 COMPUTING ARITHMETIC OPERATIONS

6.4.1 Computational problems
6.4.2 Scientific calculators
6.4.3 Mathematical tables and charts

UNIT A6.5 SOLVING PROBLEMS USING MENSURATION

6.5.1 Properties of quadrilaterals
6.5.2 Areas of plane figures
6.5.3 Volumes of regular figures

MODULE: 74-07-A DOOR AND WINDOW OPENINGS

NOMINAL DURATION: 100 HOURS CREDITS: 10

MODULE PURPOSE: To equip the trainee with knowledge, skills and appropriate attitude to effectively and efficiently fit door and window frames

LEARNING OUTCOMES: On completion of this module, trainees will be able to:

1. Construct jambs and reveals
2. Fit door frames
3. Fit window frames
4. Cast Lintel
5. Place Lintel
6. Build Threshold

UNIT A7.1 CONSTRUCTING JAMBS AND REVEALS

7.1.1 Measurements
7.1.2 Types of jambs and reveals
7.1.3 Methods of construction

UNIT A7.2 FITTING DOOR FRAMES

7.2.1 Types of door frames
7.2.2 Frames and linings
7.2.3 Materials
7.2.4 Methods of fitting

UNIT A7.3 FITTING WINDOW FRAMES

7.3.1 Types of window frames
7.3.2 Fitting principles
7.3.3 Materials
7.3.4 Methods of fitting

UNIT A7.4 CASTING AND PLACING LINTEL

7.4.1 Types of lintels
7.4.2 Formwork
7.4.3 Materials
7.4.4 Methods of construction and placing

UNIT A7.5 BUILDING THRESHOLD

7.5.1 Purposes
7.5.2 Types
7.5.3 Materials
7.5.4 Methods of construction
7.5.5 Finishing

MODULE: 74-08-B FINISHES

NOMINAL DURATION: 100 HOURS CREDITS: 10

MODULE PURPOSE: To equip the trainee with knowledge, skills and appropriate attitudes to effectively and efficiently apply wall finishes.

LEARNING OUTCOMES: On completion of this module, trainees will be able to:

1 Plaster different types of walls
2 Decorate walls and ceilings with gypsum
3 Joint different types of walls
4 Point different types of walls
5 Lay floor screeds
6 Fix tiles
7 Fix cladding on structure

UNIT B8.1 PLASTERING DIFFERENT TYPES OF WALLS

8.1.1 Rules of plastering
8.1.2 Composition and ratios
8.1.3 Types of plaster and background
8.1.4 Methods of application
8.1.5 Curing

UNIT B8.2 JOINTING DIFFERENT TYPES OF WALLS

8.2.1 Types of joints
8.2.2 Factors to consider when joining
8.2.3 Jointing Procedures

UNIT B8.3 POINTING DIFFERENT TYPES OF WALLS

8.3.1 Types of pointing
8.3.2 Advantages and disadvantages
8.3.3 Mortars and colouring agents
8.3.4 Pointing procedures

UNIT B8.4 LAYING FLOOR SCREEDS

8.4.1 Types of floor screeds
8.4.2 Methods of application
8.4.3 Curing screeds

UNIT B8.5 FIXING TILES

8.5.1 Surface preparation
8.5.2 Types of tiles
8.5.3 Adhesive materials
8.5.4 Methods of fixing

UNIT B8.6 **FIXING CLADDING ON STRUCTURES**

8.6.1 Surface preparation
8.6.2 Types of background
8.6.3 Types of cladding
8.6.4 Methods of fixing

UNIT B8.7 **DECORATING WALLS AND CEILINGS WITH GYPSUM (RHINOLITE)**

8.7.1 Surface preparation
8.7.2 Types of background
8.7.3 Methods of fixing
8.7.4 Finishing

MODULE: 74-09-B RENOVATING AND ALTERING STRUCTURES

NOMINAL DURATION: 120 HOURS CREDITS: 12

MODULE PURPOSE: To equip the trainee with knowledge, skills and appropriate attitude to effectively and efficiently communicate.

LEARNING OUTCOMES: On completion of this module, trainees will be able to:

1 Demolish structures
2. Make alterations on floors and walls
3 Repair defects on structures
4. Repair pavements and walkways
5. Paint structures

UNIT B9.1 DEMOLISHING STRUCTURES

9.1.1 Demolition tools and equipment
9.1.2 Nature of demolition
9.1.3 Demolition procedures
9.1.4 Rubble Removal

UNIT B9.2 MAKING ALTERATIONS ON FLOORS AND WALLS

9.2.1 Marking out
9.2.2 Partitioning
9.2.3 Expansion joints
9.2.4 Reinforcement of structures
9.2.5 Underpinning

UNIT B9.3 REPAIRING DEFECTS ON STRUCTURES

9.3.1 Cracking walls
9.3.2 Cracking floors
9.3.3 Tile replacement
9.3.4 Shoring
9.3.5 Timbering

UNIT B9.4 REPAIRING PAVEMENTS AND WALKWAYS

9.4.1 Surface preparation
9.4.2 Ground determination
9.4.3 Types of pavers and kerbstones
9.4.4 Laying procedures

UNIT B9.5 PAINTING STRUCTURES

9.5.1 Surface preparation
9.5.2 Materials
9.5.3 Types of paint
9.5.4 Tools
9.5.5 Painting techniques

MODULE: 74-10-B DRAINAGE

NOMINAL DURATION: 100 HOURS CREDITS: 10

MODULE PURPOSE: To equip the trainee with knowledge, skills and appropriate attitudes to effectively and efficiently construct drainage systems.

LEARNING OUTCOMES: On completion of this module, trainees will be able to:

1. Lay pipes
2. Construct surface drainage
3. Construct subsoil drainage
4. Build sewage disposal systems
5. Lay combined drainage systems

UNIT B10.1 LAYING PIPES

10.1.1 Cast iron drain pipes and traps
10.1.2 Clay drain pipes and traps
10.1.3 Asbestos-Cement drain pipes
10.1.4 Plastic pipes
10.1.5 Concrete pipes

UNIT B10.2 CONSTRUCTING SURFACE DRAINAGE

10.2.1 Disposal of rain water
10.2.2 Types of Gullies
10.2.3 Types of Soak away
10.2.4 Determination of gradient and flow velocity

UNIT B10.3 CONSTRUCTING SUBSOIL DRAINAGE

10.3.1 Application areas
10.3.2 Materials
10.3.3 Procedures of application

UNIT B10.4 BUILDING SEWAGE DISPOSAL SYSTEMS

10.4.1 Inspection chamber details
10.4.2 Manhole specifications
10.4.3 Interception chambers
10.4.4 Septic tank details
10.4.5 Soak away techniques

UNIT B10.5 LAYING COMBINED DRAINAGE SYSTEMS

10.5.1 Connections
10.5.2 Inspections
10.5.3 Disinfecting
10.5.4 General cleaning
10.5.5 Repair works

MODULE 74 -11-B ENTREPRENEURSHIP

DURATION: 80 HOURS CREDITS: 8

MODULE PURPOSE: To equip trainees with entrepreneurial knowledge, skills and appropriate attitude to run profitable enterprises efficiently

LEARNING OUTCOMES: On completion of the module, the trainees will be able to:

1. Apply entrepreneurial concepts
2. Demonstrate entrepreneurial competencies and attitudes
3. Identify business opportunities
4. Interpret the entrepreneurial environment
5. Develop business plan and establishing the business
6. Manage finances
7. Manage a business

UNIT: B11.1 APPLYING ENTREPRENEURIAL CONCEPTS

11.1.1 Concepts of entrepreneurship
11.1.2 Economic trends in Zambia
11.1.3 Role of entrepreneurship in national development
11.1.4 Informal and formal sectors
11.1.5 Types of entrepreneurs

UNIT: B11.2 DEMONSTRATING ENTERPRENEURIAL COMPETENCES

11.2.1 Identifying competences
11.2.2 Assessing competences
11.2.3 Applying competences

UNIT: B11.3 IDENTIFYING BUSINESS OPPORTUNITIES

11.3.1 Generating and selecting business ideas
11.3.2 Business opportunity
11.3.3 Sources of business ideas/information
11.3.4 Business ideas through intellectual property
11.3.5 Risk management in start-ups and selecting advisors

UNIT: B11.4 INTERPRETING THE ENTREPRENEURIAL ENVIRONMENT

11.4.1 Legal environment
11.4.2 Political environment
11.4.3 Economic environment
11.4.4 Social cultural environment
11.4.5 Ethical issues and social responsibility

UNIT: B11.5 DEVELOPING BUSINESS PLAN AND ESTABLISHING THE BUSINESS

11.5.1 Feasibility of a business idea

11.5.2 Business enterprise

11.5.3 Procedures for formalizing an enterprise

11.5.4 Methods of resource mobilization

UNIT: B11.6 MANAGING FINANCES

11.6.1 Sources of finance

11.6.2 Book keeping procedures

11.6.3 Costing and pricing of products/services

11.6.4 Business performance

11.6.5 Budgets and establishing financial controls

11.6.6 Capital management

11.6.7 Investment appraisal

11.6.8 Financial statements

UNIT: B11.7 MANAGING A BUSINESS

11.7.1 Managerial and leadership skills in an enterprise

11.7.2 Decision making skills

11.7.3 Product marketing skills

11.7.4 Business record

11.7.5 Principles of costing and pricing of production/services

11.7.6 Enterprise and technology principles

11.7.7 Business ethics and value in managing an enterprise

MODULE: 74-12-B STAIRWAYS, MASONRY AND FIREPLACES

NOMINAL DURATION: **100 HOURS** **CREDIT:10**

PURPOSE: To equip the trainee with knowledge, skills and appropriate attitudes to construct stairways, fireplace and masonry works

LEARNING OUTCOMES: Upon completion of this module, the trainee will be able to:

1. Construct fireplace
2. Construct stairways
3. Build Threshold
4. Construct decorative structures
5. Carryout masonry works

UNIT B12.1 CONSTRUCTING FIREPLACE

12.1.1 Types of fireplace

12.1.2 Positioning fireplace

12.1.3 Material for fireplace

12.1.4 Chimney construction

UNIT B12.3 CONSTRUCTING STAIRWAYS

12.2.1 Materials

12.2.2 Types of staircases

12.2.3 Techniques

UNIT B12.3 BUILDING THRESHOLD

12.3.1 Materials

12.3.2 Purposes

12.3.3 Types

12.3.4 Finishing

12.3.5 Methods of construction

UNIT B12.4 CONSTRUCTING DECORATIVE STRUCTURES

12.4.1 Decorative brickwork

12.4.2 Arches

12.4.3 Piers

12.4.4 Methods of construction

UNIT B12.5 CARRYING OUT MASONRY WORKS

12.5.1 Types of stones

12.5.2 Masonry bonds

12.5.3 Setting out

12.5.4 Laying procedures

12.5.5 Finishing

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